

**Dissertation / Project / Project Work Title:**  
Automated Document Review and Task Tracking System  
Using Python-Based Workflow Automation

**Course No.:** SEZG628T

**Course Title:** Dissertation / Project / Project Work

**Dissertation / Project /Project Work Done by:** V HARSHITHA

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**BITS ID:** 2024TM93089

**Degree Program:** MTECH IN SOFTWARE  
ENGINEERING

**Research Area:** Software Engineering

**Dissertation / Project Work carried out at:**  
Accenture, Bangalore



**BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE,  
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## 1. Broad Area of Work

Software Engineering, Workflow Automation and Full-Stack Application Development

The main goal of this project is to design and develop a secure and automated Full-stack application to manage documents and track tasks that helps to handle workflows more efficiently than in a traditional application which rely on manual monitoring. The implementation mainly focuses on automating document compliance checks, applying role-based access control to ensure data security and keeping a clear audit trail of all system activities for transparency.

## 2. Background

In highly regulated fields such as life sciences, reviewing and approving documents is not just a routine administrative activity, it plays a major role in maintaining compliance, accuracy and overall data integrity. However, many organizations still rely on manual and error-prone processes or on rigid enterprise tools that are difficult to adapt. These traditional systems often act like black boxes, offering little transparency and limiting flexibility when workflow requirements change.

The current existing document management solutions depend on pre-configured workflow engines where customization is restricted by fixed rules and closed designs. As a result, engineers find it difficult to understand how the system works internally or to modify and improve its behavior. This highlights the need for a more open and flexible approach that clearly shows how document lifecycle management and workflow automation can be built and controlled from the ground up.

This project aims to address this challenge by developing an automated document review and task-tracking system. The solution combines a custom Python-based automation engine with a full-stack web application to create a flexible and transparent framework. Through this approach, the project demonstrates that document lifecycle management can be both efficient and easy to understand when implemented using modern software engineering practices.

## 3. Objectives

The primary objectives of this project are:

1. To design and develop an automated document review system with clearly defined workflow states.
2. To implement a custom workflow engine that controls document lifecycle transitions based on user roles and actions.
3. To automate task creation, assignment and closure during document review and approval processes.
4. To provide a user-friendly web interface for document management, task handling and workflow visualization.
5. To maintain an audit log for all system actions to ensure traceability and

accountability.

6. To demonstrate end-to-end full-stack application development using Python and React.

#### 4. Scope of Work

- User authentication and role-based access control (Author, Reviewer, Approver).
- Document upload and metadata management.
- Workflow states such as Draft, In Review, Approved and Rejected.
- Rule-based workflow transitions triggered by user actions.
- Automated task creation and assignment.
- Dashboard for system overview and metrics.
- “My Tasks” interface for pending workflow actions.
- Audit log captures all document and workflow activities.
- Backend implementation using Python (FastAPI).
- Frontend implementation using React.
- Advanced machine learning or AI-based document analysis as a Enhancement feature.

#### 5. Plan of Work

| Phases                                                   | Start Date-End Date           | Work to be done                                                                 |
|----------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------|
| Dissertation Outline                                     | 17 Jan 2026 – 31 Jan 2026     | Literature Review and prepare Dissertation Outline                              |
| Design & Development of MVP                              | 01 Feb 2026 – 20 Mar 2026     | Design & Development Activity                                                   |
| Development of Additional Enhancement features & Testing | 21 Mar 2026 – 10 Apr 2026     | Developing additional features & Software Testing                               |
| Dissertation Review                                      | 11 April 2026 - 30 April 2026 | Submit Dissertation to Supervisor & Additional Examiner for review and feedback |

|            |                           |                                             |
|------------|---------------------------|---------------------------------------------|
| Submission | 01 May 2026 - 30 May 2026 | Final Review and submission of Dissertation |
|------------|---------------------------|---------------------------------------------|

## 6. Literature References

1. Sommerville, I., *Software Engineering*, Pearson Education.
2. Pressman, R. S., *Software Engineering: A Practitioner's Approach*, McGraw-Hill.
3. Fowler, M., *Patterns of Enterprise Application Architecture*, Addison-Wesley.
4. W3C, *RESTful Web Services Architecture*, W3C Recommendations.
5. Richards, M., & Ford, N., *Fundamentals of Software Architecture*, O'Reilly Media.
6. IEEE, *Software Architecture and Design Standards*, IEEE Publication.

## 1. Particulars of the Supervisor and Examiner

|                                     | Supervisor                                 | Additional Examiner           |
|-------------------------------------|--------------------------------------------|-------------------------------|
| Name                                | Priyanka Nagane                            | Sandeep Parakkunath           |
| Qualification                       | B.E.                                       | MCA                           |
| Designation                         | Packaged Application Development Team Lead | Offshore Delivery Manager     |
| Employing Organization and Location | Accenture, Pune                            | Accenture, Bangalore          |
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## 2. Remarks of the Supervisor

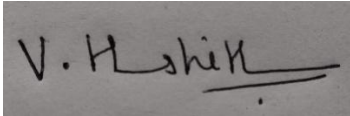
The study and project appear to be very promising. The student has covered all the relevant literature in a comprehensive manner. The project aligns well with the current research trends in the domain and meets the expected standards of a master's level dissertation. It also includes a user study conducted on the working prototype, which effectively validates the research work. Additionally, the project has the potential for future publication. I approve the project in my capacity as the supervisor.

**BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI**  
**WORK INTEGRATED LEARNING PROGRAMMES (WILP) DIVISION**  
**SECOND SEMESTER OF ACADEMIC YEAR 2021-2022**

SEZG628T: Dissertation / Project / Project Work OUTLINE

|                                                                    |                                                                                           |
|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>STUDENT ID No.</b>                                              | 2024TM93089                                                                               |
| <b>NAME OF THE STUDENT</b>                                         | V HARSHITHA                                                                               |
| <b>STUDENT'S EMAIL ADDRESS</b>                                     | <a href="mailto:harshithav99@gmail.com">harshithav99@gmail.com</a>                        |
| <b>STUDENT'S EMPLOYING ORGANIZATION &amp; LOCATION</b>             | Accenture, Bangalore                                                                      |
| <b>SUPERVISOR'S NAME</b>                                           | Priyanka Nagane                                                                           |
| <b>SUPERVISOR'S EMPLOYING ORGANIZATION &amp; LOCATION</b>          | Accenture, Pune                                                                           |
| <b>SUPERVISOR'S EMAIL ADDRESS</b>                                  | priyanka.nagane@yahoo.in                                                                  |
| <b>ADDITIONAL EXAMINER'S NAME</b>                                  | Sandeep Parakkunath                                                                       |
| <b>ADDITIONAL EXAMINER'S EMPLOYING ORGANIZATION &amp; LOCATION</b> | Accenture, Bangalore                                                                      |
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| <b>DISSERTATION / PROJECT / PROJECT WORK TITLE</b>                 | Automated Document Review and Task Tracking System Using Python-Based Workflow Automation |

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|  |                                |                                         |
| <b>Signature of Student</b>                                                         | <b>Signature of Supervisor</b> | <b>Signature of Additional Examiner</b> |
| <b>Name:</b> V Harshitha                                                            | <b>Name:</b> Priyanka Nagane   | <b>Name:</b> Sandeep Parakkunath        |